

# Simone Vellei

Senior Backend Engineer | Go · Distributed Systems · IIoT · AI Agents | Open-Source Author

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## PROFESSIONAL SUMMARY

Senior backend engineer with 20+ years of experience building software from the protocol layer up, from embedded C stacks to distributed systems and IoT platforms at the scale of millions of resources. Open-source author of Go frameworks for LLM and multi-agent systems. Works AI-first with Claude Code and spec-driven development; builds small, high-ownership teams.

## EXPERIENCE

Senior Backend Engineer — Cybus

04/2024 - Present

Remote, Italy · Tech stack: Go, Node.js, NATS, Kubernetes, PostgreSQL  
Cybus Connectware is an Industrial IoT platform for factory data integration.

- Redesigned the resource status tracking subsystem of Connectware in Go, replacing a bottlenecked component that was limiting platform scalability; the new control plane lifted the ceiling to 1 million resources.
- Co-designed the migration from a synchronous RPC model to a NATS-based event-driven architecture for inter-service communication, driving key algorithmic and design decisions within a 3-person architecture team; building on the new control plane, this raised capacity to 3 million resources, a 100x improvement over the original platform.
- As the sole engineer with hands-on generative AI experience at Cybus, designed and delivered a company-wide training on LLMs, embeddings, RAG, vector databases, and multi-agent architectures, driving team-wide adoption of AI-native development practices, including spec-driven Claude Code workflows.
- Authored the architecture decision record and led a small engineering team through the design and delivery of Connectware's monitoring feature, from technical approach through rollout; the feature was a requirement for a new customer and directly secured the deal.
- Implemented and shipped to production an MCP server exposing Connectware's capabilities to AI agents, enabling agent-driven interaction with the platform.

Senior Backend Engineer — UFirst

11/2021 - 04/2024

Remote, Italy · Tech stack: Go, Python, Kubernetes, GCP, AWS, MongoDB  
UFirst is a SaaS platform for digital queue management, virtual appointments, and customer flow analytics.

- Engineered concurrent Go microservices, deployed on Kubernetes across GCP and AWS, offloading in-application aggregation logic to MongoDB pipelines to reduce service memory footprint under high booking load.
- Led the architectural decision to adopt Amazon Chime SDK over WebRTC for video-conference support, avoiding distributed signaling complexity; clients could book, manage, and conduct appointments without leaving the platform, supporting services such as remote bank mortgage consultations.
- Designed and implemented in Python the multi-service worker pipeline for ExtraI, a private-LLM SaaS platform for enterprise process automation, applying content-aware chunking strategies per input type to maximize LLM input quality; the system processed hundreds of conference speeches across audio, video, and slide inputs, generating per-session chatbots for attendees.

Backend Engineer / Tech Lead — Inim Electronics

01/2017 - 11/2021

Monteprandone (AP), Italy · Tech stack: Go, MySQL, Docker, DigitalOcean, AWS

- Built the company's cloud engineering function from scratch: hired and grew a team of 3, defined project management processes and code quality standards, provided direct architectural mentoring, and contributed hands-on to implementation throughout.
- Re-implemented the proprietary communication protocol stack and re-engineered the TCP/IP layer of resource-constrained burglar-alarm panels to enable cloud connectivity, keeping memory and CPU footprint within tight embedded system limits.
- Architected a cloud infrastructure to connect and manage hundreds of thousands of alarm panels deployed worldwide, designing for horizontal scalability and high device-connection density, and leading an external team through implementation.
- Researched Google Assistant and Amazon Alexa integration protocols, designed the bridge to the existing proprietary Inim cloud API, and guided the external development team through implementation to ensure protocol compliance and system stability.
- Guided a firmware team through their first embedded TCP/IP implementation: researched and selected suitable networking libraries for constrained hardware, defined the integration approach, and contributed hands-on to development alongside the team.

Monteprandone (AP), Italy · Tech stack: C, Embedded Linux, Buildroot

- Engineered a proprietary secure internet protocol for a new generation of burglar-alarm devices, enabling home automation integration and more complex system topologies.
- Delivered the company's first GNU/Linux embedded LAN gateway for burglar alarms, adding support for third-party protocols (KNX, SIA-IP, Modbus), a responsive web interface, and email notifications.
- Implemented a token-ring network stack over RS-485 that allowed up to 30 fire-detection panels to operate as a single networked system, large enough to cover a major airport.

Junior Linux Embedded Engineer — Bentel Security

04/2005 - 09/2005

Corropoli (TE), Italy

- Built the first prototype of a GNU/Linux-based multimedia keypad for burglar alarms and authored the product specifications for the resulting product line.

OPEN SOURCE PROJECTS

LinGoose    Go framework for building AI/LLM applications

<https://github.com/henomis/lingoose> | <https://simonevellei.com/lingoose>

- Designed and built a modular, abstraction-based Go framework for LLM application development (829+ GitHub stars, 75 forks).
- Integrates with OpenAI, Anthropic, and other LLM providers; covers embeddings, vector search, prompt pipelines, and custom extensions.
- Enabled developers to selectively import only needed components, reducing dependency footprint.

Phero    Modern Go framework for multi-agent AI systems

<https://github.com/henomis/phero> | <https://simonevellei.com/phero>

- Designed and built a composable multi-agent orchestration framework with runtime agent handoffs and MCP integration (60+ GitHub stars).
- Built-in RAG with Qdrant, PostgreSQL/pgvector, and Weaviate backends; Agent-to-Agent (A2A) protocols for inter-process communication.
- Typed observability via lifecycle event tracing, and LLM middleware for cross-cutting behaviors.

Full portfolio of public projects:    <https://github.com/henomis>

EDUCATION

Master of Science (M.Sc.) in Computer Science

2005

University of Bologna (Università di Bologna)

SKILLS

**Languages:** Go (Golang) | JavaScript / Node.js | Python | C (embedded)

**Generative AI:** Large Language Models (LLMs) | LLM API integration | Embeddings | Vector databases | Retrieval-Augmented Generation (RAG) | Multi-agent architectures | AI agents | Agent-to-Agent (A2A) | MCP (Model Context Protocol) servers | OpenAI & Anthropic APIs | Prompt engineering

**AI-Assisted Development:** Claude Code | Spec-driven development | Agentic coding workflows | Custom skills & subagents | CLAUDE.md / context engineering

**Architecture:** Distributed systems | Microservices | Event-driven architecture | IoT / IIoT | Embedded Linux

**Leadership:** Technical leadership | Mentoring | Hiring | Architecture decision records (ADR)

**Infrastructure & Protocols:** NATS | MQTT | REST APIs | GraphQL | Docker | Kubernetes | AWS | GCP | DigitalOcean | CI/CD

**Databases:** MongoDB | Redis | MySQL | PostgreSQL/pgvector | Qdrant | Weaviate

**Observability:** Prometheus | Grafana